

Sparse Feature Selection from Low-Montage Resting-State EEG for tDCS Treatment Remission Prediction in Adults with Major Depressive Disorder

Abin Daniel Zorto^{1*}, Jijomon C. Moncy², Wenyi Xiao²,
Mhd Saeed Sharif¹, Cynthia H.Y. Fu²

¹Intelligent Technologies Research Group, University of East London, University Way, London E16 2RD, UK

²School of Psychology, University of East London, University Way, London E16 2RD, UK

Abstract

This study evaluated a sparse low-montage resting-state EEG framework for identifying clinically interpretable features associated with remission outcomes in a home-based transcranial direct current stimulation (tDCS) randomised sham-controlled trial in adults with major depressive disorder. Using 4-channel eyes-closed EEG, we conducted a controlled sweep of 150 successful runs across two model families, an advanced hybrid 1D CNN-LSTM and a linear support vector machine (SVM), varying window size (2–10 s) and per-fold feature budget (**inner-k** 1–70) with leave-one-participant-out group equalization and SMOTE during training.

The strongest configuration (hybrid model, 6 s windows, **inner-k=1**) achieved patient-level ROC-AUC 0.816, PR-AUC 0.579, balanced accuracy 0.786, accuracy 0.810, and MCC 0.571 over 21 held-out participants. The permutation test confirmed a significant advantage over chance (raw $p = 0.008$), which survives Bonferroni correction for the two model-family comparisons ($p_{\text{corr}} = 0.016$). As robustness checks, a max-T permutation (10,000 iterations across all 150 sweep configurations) yielded $p_{\text{corr}} = 0.302$, and the full-sweep Bonferroni was mechanically uninformative ($p_{\text{corr}} = 1.00$) given the 150-test multiplier; we report all three tiers. The best SVM configuration (8 s, **inner-k=30**) reached ROC-AUC 0.510 and MCC 0.277 ($p = 0.469$, uncorrected). Notably, the best hybrid performance was obtained with extreme sparsity, selecting 1 feature per fold and yielding only 7 unique features in at least one fold. Hard-set overlap across folds was modest (mean pairwise Jaccard 0.190; Kuncheva index 0.187), while recurrent features showed perfect mean effect-direction consistency.

These findings suggest that patient-level discrimination signals may be recoverable from highly sparse representations of low-montage EEG, although statistical significance does not survive correction for the full sweep. The strongest run concentrated its signal in a small recurrent feature set centered on temporal spectral entropy and frontal-temporal difference measures. The results converge with previous power spectral density-based findings and support low-montage resting-state EEG as a basis for further pretreatment stratification research, pending independent validation.

*Corresponding author: Abin Daniel Zorto, u2091940@uel.ac.uk

1 Introduction

Major depressive disorder (MDD) remains a leading cause of global disability and is characterized by marked heterogeneity in clinical presentation, neurobiology, and treatment response. Despite the availability of pharmacological and psychotherapeutic interventions, a substantial proportion of patients fail to achieve remission, motivating increasing interest in neuromodulation approaches such as transcranial direct current stimulation (tDCS) [?]. While randomized controlled trials and recent large-scale studies support the feasibility and clinical potential of tDCS, including fully remote, home-based delivery, treatment outcomes remain variable at the individual level [? ?].

1.1 Clinical significance of low-density EEG

Home-based delivery has moved from concept to clinical trial: a fully remote phase 2 randomized sham-controlled study demonstrated home-based tDCS in major depressive disorder [?], and qualitative follow-up work found the approach acceptable to participants [?]. These advances make pretreatment stratification a practical clinical problem. A central challenge is pretreatment stratification: identifying which patients are more likely to achieve remission before treatment begins. Electroencephalography (EEG) is a potential modality for this purpose due to its portability, low cost, and high temporal resolution. A 4-channel low-montage configuration (AF7, AF8, TP9, TP10) is particularly attractive for home-based settings: the headset is lightweight, self-applicable by patients with minimal training, and substantially less expensive than high-density laboratory systems. However, this reduced channel count necessarily trades off spatial coverage (capturing only bilateral frontal and temporal regions) and signal richness against practical deployability. A key open question is whether the limited spatial sampling of a 4-channel montage retains sufficient information for clinically useful treatment-response prediction, or whether the signal loss relative to 32- or 64-channel systems is prohibitive. The present study directly addresses this tension by testing whether rigorous feature engineering and sparse selection can recover predictive signal from the same low-density setup used in the parent home-based trial.

The literature on EEG biomarkers in major depressive disorder is encouraging though methodologically uneven. Reviews and biomarker overviews have repeatedly highlighted spectral, asymmetry, vigilance, entropy, and complexity signatures as potentially informative marker families [? ? ?]. Related discussions of neuroimaging and EEG biomarkers have emphasized the clinical opportunity, but also the need for more robust treatment-response models and better translational discipline [? ? ?]. More recent meta-analytic work on EEG-based treatment prediction reaches a similar conclusion: useful signal is present, but translation depends on stronger methodology and standardization [? ? ?]. Machine learning studies have extended this literature into antidepressant, placebo, and neuromodulation response prediction, while also exposing recurrent concerns about external generalization, feature stability, leakage-safe selection, and interpretability [? ? ?]. Related nonlinear and entropy-oriented studies point to the same caution about dimensionality and signal instability [? ? ?]. Entropy-focused EEG work has reinforced the relevance of complexity measures [? ? ?]. Leakage-specific and data-centric critiques make the underlying design problem explicit [? ?].

That tension is especially sharp in relatively small neuromodulation studies. Rich models can improve representation learning, but relatively small cohorts also increase the risk of unstable feature selection, inflated performance estimates, and overly confident biomarker claims. Nested evaluation and within-fold processing are therefore central design issues rather than implementation details in this literature [? ?]. Clinically deployable EEG biomarkers also need interpretability and low-density feasibility, and recent data-centric frameworks have started to move in that direction [?

]. Emerging response-prediction studies in depression and related neuromodulation settings show substantial promise [? ? ?]. Related reviews and pilot studies also make the methodological fragility clear [? ? ?]. More recent response-oriented studies continue that pattern [? ? ?]. A recent systematic review of EEG connectivity changes in depression treatment reaches a similar conclusion [?].

Within the same home-based 4-channel MDD program analysed here, power spectral density (PSD)-based and PLV-based analyses have observed that the baseline EEG carries treatment-response information across multiple signal representations [? ?]. PSD-based deep learning approaches identified temporoparietal spectral features, notably TP10 theta-alpha power, as informative for treatment outcome prediction [?], while connectivity-based analyses highlighted frontal-temporal and temporoparietal network patterns [?]. Together, these findings suggest that response-related signal is present and reproducible across feature families, but remains poorly characterized in terms of interpretability, stability, and minimal feature representation.

The present study builds directly on this context and uses the same baseline EEG cohort as our prior PSD- and connectivity-based analyses [? ?], but addresses a distinct question. Rather than optimising predictive performance within a predefined feature family, we test whether a strictly leakage-controlled, foldwise sparse feature-selection pipeline can identify a small, recurrent, and interpretable feature set under participant-level validation. We compared two classifiers over the same feature-selection sweep to test whether low-montage resting-state EEG could recover a small recurring set of features associated with tDCS remission in adults with MDD. The analysis prioritized parsimony, interpretability, and leakage-resistant participant-level validation. Our working hypothesis was that useful patient-level discrimination could be achieved with a small per-fold feature budget and that the most recurrent features would show stronger effect-direction consistency than hard-set overlap. This approach reframes the problem from high-dimensional modelling toward a minimal, interpretable signal identification task.

2 Methods

2.1 Dataset and preprocessing

The analysis used pretreatment EEG from 21 adults with major depressive disorder enrolled in a fully remote, multisite, double-blind, randomized sham-controlled home-based tDCS trial [?]. The analyzed cohort contained 7 remission cases and 14 non-remission cases in the patient-level outcome labels, with 18 female participants and a mean age of 37.1 ± 9.7 years. Table 1 summarizes the cohort, acquisition, and sweep configuration.

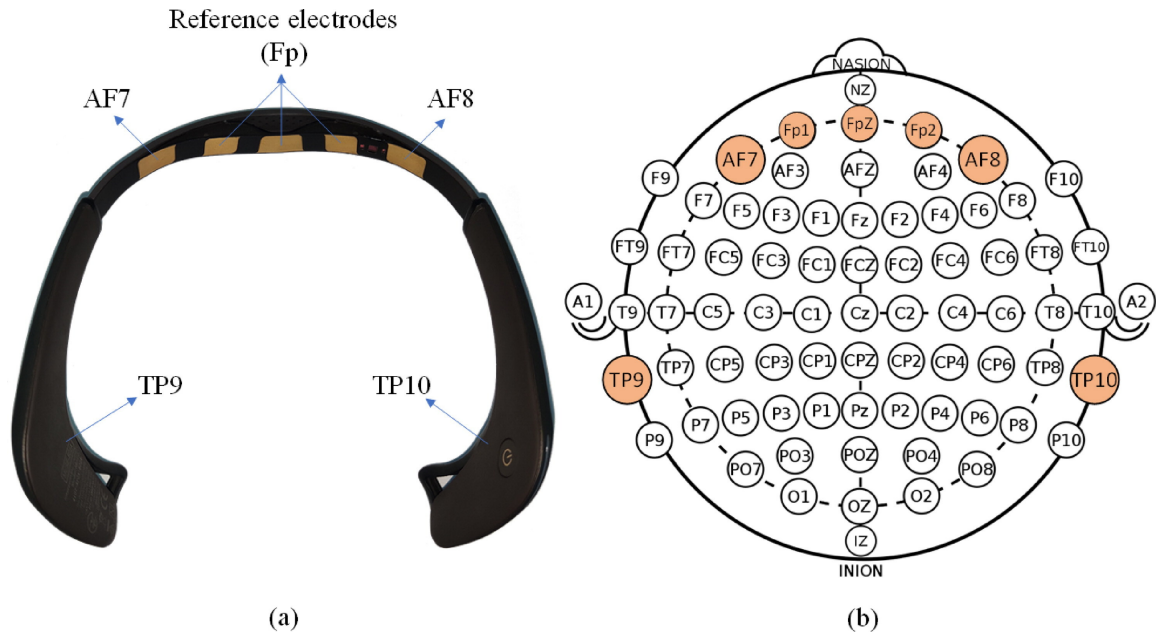


Figure 1: Four-channel low-montage EEG setup corresponding to the recording configuration used in the study, showing the AF7, AF8, TP9, and TP10 recording sites together with the corresponding headset placement.

Table 1: Cohort, acquisition, and sweep configuration summary. Trial descriptors derive from the parent home-based tDCS study [?]; acquisition totals and sweep settings derive from the analyzed EEG exports and current experiment configuration.

Characteristic	Value
Parent trial and cohort descriptors	
Parent trial	Fully remote, multisite, sham-controlled home-based tDCS trial for MDD [?]
Participants	21
Outcome groups	7 remission, 14 non-remission
Sex (female/male)	18/3
Age (years, mean $\pm$ SD)	37.1 $\pm$ 9.7
EEG acquisition and export descriptors	
Rest condition	Eyes-closed resting state
EEG montage	4 channels (AF7, AF8, TP9, TP10)
Channel coverage	Frontal (AF7, AF8), temporal (TP9, TP10)
Sampling rate	256 Hz
Recording structure	10 min total; single file or two 5 min files
Available export segment length	10 s (2,560 samples/channel)
Total exported windows	1,203
Remission windows	393 (32.7%)
Non-remission windows	810 (67.3%)
Class ratio	2.06:1 (non-remission:remission)
Current sweep methodology settings	
Sweep window sizes	2, 4, 6, 8, 10 s
Inner-k sweep	1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70
Feature selector	SelectKBest (f -classif)
Training adjustments	LOPO-group equalization + SMOTE
Validation	Leave-one-participant-out
Evaluation level	Patient-level aggregation

Baseline recordings were collected during eyes-closed rest with a 4-channel montage (AF7, AF8, TP9, TP10) sampled at 256 Hz. Figure 1 shows the recording sites together with the corresponding headset placement. The available study exports comprised 10 s segments with 2,560 samples per channel, and the underlying recording sessions were stored either as a single 10 min file or as two 5 min files depending on participant. Across the cohort, the export set contained 1,203 windows, including 393 remission-labeled and 810 non-remission-labeled windows. This acquisition setup uses a limited frontal-temporal montage with clear portable-deployment relevance [?]. The montage also constrained the downstream feature space toward frontal, temporal, cross-hemispheric, and frontal-temporal interaction measures.

For the present sweep, the processing configuration applied participant-level per-channel demeaning before upsampling, followed by a fourth-order Butterworth low-pass filter at 60 Hz, a downsampling step back to 256 Hz, and zero-overlap window slicing. The 60 Hz cutoff was chosen to retain the full gamma band while providing attenuation above the Nyquist frequency of the original 256 Hz recording; a notch filter at 50 Hz (UK mains frequency) was not applied because the low-pass rolloff already provides substantial attenuation. The upsample/downsample pipeline preserved signal fidelity while allowing consistent filter implementation across all window sizes. The sweep then compared non-overlapping analysis windows of 2, 4, 6, 8, and 10 s, so the model inputs included both the native 10 s exports and shorter subdivisions derived from the same recordings.

These choices materially affect downstream EEG feature distributions, especially in low-density or portable pipelines [? ? ?]. They also intersect with standardization and signal-quality concerns in wearable EEG settings [? ?].

2.2 Feature extraction

Feature extraction was performed on each preprocessed window. With all four channels available, the extractor generated 247 EEG-derived measures per window spanning channel-local descriptors, regional asymmetry terms, frontal-temporal interaction terms, synchrony summaries, and a small set of TP10-specific derived measures. Each channel contributed 39 local measures, comprising 21 spectral descriptors, 12 time-domain and Hjorth descriptors, and 6 entropy or nonlinear complexity descriptors, for 156 channel-local measures overall. The remaining 91 measures summarized asymmetry, frontal-temporal balance, cross-channel synchrony, and TP10-focused derived descriptors. This broad feature basis allowed multiple interpretable feature families to be tested within a leakage-safe sparse-selection framework rather than pre-committing the analysis to a single biomarker hypothesis [? ?]. Table 2 summarizes the extracted feature families and their computational rationale.

For each of AF7, AF8, TP9, and TP10, we computed 39 channel-local features. Spectral features were estimated from Welch power spectral densities [?] and included absolute band power in the delta, theta, alpha, beta, and gamma ranges, total power, relative band power normalized by total power, frequency-weighted power, spectral edge frequencies at 90% and 95%, and PSD summary statistics (mean, standard deviation, maximum, minimum, median, skewness, kurtosis). These measures were included because spectral content remains one of the most widely used and interpretable feature families in depression EEG work [? ? ?]. Temporal-channel spectral content has also appeared repeatedly in pretreatment response studies [?].

Table 2: Summary of extracted EEG feature families per analysis window.

Feature family	Measures	Count	Why included
Channel-local spectral	Absolute and relative delta, theta, alpha, beta, and gamma power; total power; frequency-weighted power; SEF90 and SEF95; PSD mean, standard deviation, maximum, minimum, median, skewness, and kurtosis	21/channel (84 total)	Canonical resting-state EEG descriptors with clear band-level interpretation and strong precedent in depression EEG
Channel-local time domain and Hjorth	Mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, mean absolute deviation, Hjorth activity, mobility, and complexity	12/channel (48 total)	Captures amplitude dispersion, oscillatory rate, roughness, and short-window morphology with low computational cost
Channel-local entropy and nonlinear complexity	Sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis	6/channel (24 total)	Captures irregularity and complexity beyond band-power summaries
Band-power asymmetry and frontal-temporal interactions	Frontal and temporal asymmetry, hemispheric asymmetry, ipsilateral frontal-temporal ratios and normalized differences, and diagonal AF7-TP10 and AF8-TP9 interactions across frequency bands	70 total	Tests laterality and frontal-temporal balance within the four-channel montage
Synchrony and coupling	Pearson correlation and maximum normalized cross-correlation over six anatomical channel pairs, plus legacy cross-correlation summaries	18 total	Summarizes short-window coupling and cross-channel coordination
TP10-specific derived descriptors	Peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term	3 total	Targets concentrated temporal spectral structure with straightforward interpretation
Total EEG-derived features	Combined feature space per window	247 total	Broad hypothesis space for leakage-safe sparse selection

Time-domain descriptors were also computed for each channel. These included mean, standard deviation, variance, skewness, kurtosis, root-mean-square amplitude, peak-to-peak range, zero crossings, and mean absolute deviation, together with Hjorth activity, mobility, and complexity [? ?]. These features summarize amplitude dispersion, local oscillatory rate, signal roughness, and short-window morphology. They are computationally light and well suited to low-montage resting-state EEG, where clinically useful features must remain interpretable and robust under short windows.

Entropy and nonlinear complexity measures were computed per channel to capture irregularity beyond simple power summaries. The extractor included sample entropy, spectral entropy, correlation dimension, Hurst exponent, Lyapunov exponent, and detrended fluctuation analysis. Complexity measures were calculated on a demeaned, standardized version of each signal, with long windows subsampled for numerical stability. These descriptors were retained because depression EEG studies have repeatedly implicated entropy, variability, and broader complexity structure [? ? ?]. That pattern also appears in pretreatment response and large-scale resting-state evaluations, even when the exact direction of effect depends on cohort, montage, and temporal scale [? ? ?].

The four-channel montage also supports a range of regional interaction features derived from

cached band-power estimates. Traditional frontal asymmetry (AF7/AF8) and temporal asymmetry (TP9/TP10) were computed for each frequency band in two forms: a log-ratio index,

$$\text{AI} = \log\left(\frac{P_R}{P_L}\right),$$

and a normalized difference,

$$\text{ND} = \frac{P_R - P_L}{P_R + P_L}.$$

Using the same band-power cache, we also computed hemispheric asymmetry by averaging AF7+TP9 and AF8+TP10 within each hemisphere, ipsilateral frontal-temporal ratios and normalized differences on the left and right hemispheres, and diagonal frontal-temporal interactions for AF7-TP10 and AF8-TP9. These measures were included because reduced-channel depression EEG studies repeatedly point to laterality, temporal involvement, and frontal-temporal balance rather than to a single isolated electrode [? ? ?].

Pairwise synchrony features were added for six anatomically meaningful channel pairs: AF7-AF8, TP9-TP10, AF7-TP9, AF8-TP10, AF7-TP10, and AF8-TP9. For each pair we computed maximum normalized cross-correlation and Pearson correlation, and the extractor also retained a legacy set of raw cross-correlation summaries across all channel pairs. These features were intended to capture short-window synchrony and coupling patterns that have been emphasized in resting-state depression EEG and neuromodulation response work [? ? ?].

Finally, the extractor included three exploratory TP10-specific derived descriptors: peak prominence, peak concentration ratio, and an asymmetry-weighted peak-prominence term. These summarize whether the TP10 spectrum contains a sharp dominant peak, how concentrated that peak is relative to total power, and whether that concentration covaries with ipsilateral frontal-temporal beta balance. They were included because temporal-channel spectral content has recurrently appeared in low-montage depression EEG analyses, while still remaining interpretable enough to survive foldwise sparse selection if informative [? ?].

2.3 Sweep design and validation

The experiment sweep compared two model families over the same feature-selection protocol: an advanced hybrid 1D CNN-LSTM and a linear SVM. All runs used sequential window ordering, SelectKBest with ANOVA F statistics, leave-one-participant-out evaluation, leave-one-participant-out training-group equalization, and SMOTE within training folds [?]. The sweep varied the per-fold feature-selection budget (**inner-k**) across 15 values from 1 to 70 and repeated the search across 5 window sizes, yielding 150 successful runs in total (75 per model).

This design matters for interpretation. Leave-one-participant-out evaluation remained subject independent throughout, so no participant contributed both training and test windows to the same fold. Feature selection, participant-group equalization, and synthetic oversampling were all restricted to the corresponding training participants, which is important because leakage at these steps can materially inflate reported biomarker performance [?]. The **inner-k** parameter controlled how many features were selected inside each training fold. The feature-selection stability panels in the study describe shared pipeline behavior because both classifiers consumed the same selected features for matched sweep settings.

Resting-State EEG Processing and Evaluation Pipeline

In each fold, one participant is held out once; rebalancing and feature selection are performed only on the remaining training participants.

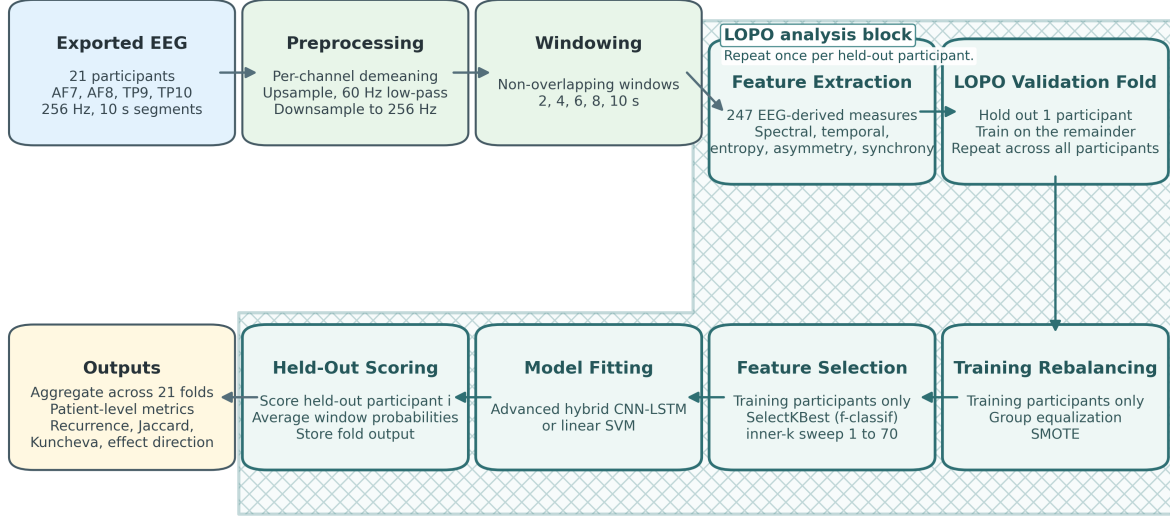


Figure 2: End-to-end analysis flow used in the study. In each leave-one-participant-out fold, one participant was held out once, and participant-group equalization, SMOTE, and feature selection were all applied only within the remaining training participants before held-out scoring and biomarker summarization.

2.4 Models

Both classifiers operated on the fold-selected handcrafted feature vectors produced by the current **inner-k** setting. Patient-level outputs were derived by averaging held-out window probabilities within each participant before computing ROC-AUC and the other clinical metrics. Table 3 summarizes the model architecture and training setup used in the sweep.

Table 3: Model architecture and training summary for the two evaluated classifier families.

Component	Configuration
Input to both models	Fold-selected handcrafted feature vector from the current <code>inner-k</code> setting. Patient-level predictions were computed by averaging held-out window probabilities within each participant.
SVM baseline	Linear-kernel SVM with $C=1.0$ and probability estimates enabled. When SMOTE was active in the training folds, class weighting was disabled to avoid double balancing.
Hybrid input embedding	Linear projection from the selected feature vector to 128 latent units, followed by normalization, a nonlinear activation, dropout 0.1, and remapping to a learned one-dimensional sequence of length 64.
CNN trunk	Three 1D convolutional blocks with filter progressions <code>[64,64]</code> , <code>[128,128]</code> , and <code>[256,128]</code> ; kernel sizes <code>[5,3]</code> , <code>[3,3]</code> , and <code>[3,1]</code> ; max-pooling factors 2, 2, and 1; batch normalization; and dropout <code>[0.1, 0.2, 0.3]</code> .
Recurrent stage	Two bidirectional LSTM layers with 128 and 64 hidden units followed the CNN trunk.
Attention and fusion	Eight-head self-attention with dropout 0.1 and residual layer normalization, followed by global mean pooling. Multi-scale pooled CNN features were concatenated through feature-pyramid fusion.
Dense head	Two dense blocks of 256 and 128 units with batch normalization, nonlinear activations, and dropout 0.3 and 0.4, followed by a two-logit output layer.
Optimization	AdamW with learning rate 0.001 and weight decay 0.01; cosine warm restarts to a minimum learning rate of 10^{-6} ; batch size 8192 (upper bound, reduced to the full per-fold training set of approximately 2,300 windows for the 6 s configuration by the batch-resolution logic, resulting in full-batch gradient descent per fold); label smoothing 0.05; gradient clipping at 1.0; mixed precision; maximum 100 epochs; early stopping patience 5 with best-state restore.

The SVM baseline used a linear kernel with $C=1.0$ and probability output enabled. When SMOTE was active inside the training folds, class weighting was not applied in parallel, so synthetic oversampling and explicit class weighting were not used simultaneously.

The hybrid model treated the selected feature vector as a learned one-dimensional representation before recurrent modeling. In the best-performing 6 s, `inner-k=1` run, the input features were first projected to a 128-dimensional embedding with normalization and dropout, then mapped into a 64-position latent representation for three convolutional blocks. These blocks used channel progressions of `[64,64]`, `[128,128]`, and `[256,128]` with kernel sizes `[5,3]`, `[3,3]`, and `[3,1]`, together with batch normalization, max pooling, and dropout after each stage.

Sequence modeling then used two bidirectional LSTM layers with 128 and 64 hidden units, followed by eight-head self-attention with residual layer normalization and dropout. The recurrent summary was globally pooled and concatenated with pooled multi-scale CNN outputs through feature-pyramid fusion before two dense blocks of 256 and 128 units and a final two-logit output layer. Training used AdamW, cosine warm restarts, label smoothing, gradient clipping, mixed precision, and early stopping with best state restore.

The architectural choices were intended to distribute the modeling burden across complementary stages. The projection layer compressed the sparse selected features into a stable latent representation before deeper processing. The convolutional front end summarized short-range interactions and produced progressively richer multi-scale feature maps. The bidirectional LSTM layers then provided recurrent processing after the convolutional front end, while self-attention allowed the model to reweight informative positions after recurrent encoding. Feature-pyramid fusion retained both shallower and deeper convolutional summaries, which is useful when the most discriminative structure may not reside at a single representational depth.

The dense head provided a final nonlinear compression stage before binary classification. Batch normalization, dropout, label smoothing, gradient clipping, and early stopping were included to stabilize optimization and limit overfitting in a relatively small cohort.

The SVM was selected as a transparent, well-characterized linear baseline, allowing direct comparison of whether a nonlinear deep architecture could extract complementary interactions from the same handcrafted features. The hybrid CNN-LSTM was chosen because it distributes representational capacity across convolutional, recurrent, and attention-based stages, making it suited to learning interactions among a small set of selected features. All hyperparameters (layer sizes, learning rate, dropout schedule, kernel sizes, attention heads, optimizer settings) were fixed a priori based on prior work with similar EEG feature dimensions and were not tuned on the current cohort; the values were inherited from a standard configuration used across multiple model families in this pipeline. No hyperparameter optimization was conducted within the present dataset, so the reported values do not reflect dataset-specific tuning. The **inner-k** and window-size sweep represents the only structured hyperparameter exploration in this study. When **inner-k=1**, SMOTE operates in a 1-dimensional feature space where synthetic interpolation reduces to jittering on a single axis; we note that the standard motivation for SMOTE does not fully apply in this regime, and we report the sensitivity of this configuration in the discussion.

2.5 Outcome metrics and feature summaries

The primary endpoint was patient-level ROC-AUC, computed from participant probabilities aggregated over held-out windows. Secondary patient-level metrics included PR-AUC, balanced accuracy, accuracy, F1, MCC, precision, recall, and specificity. The pipeline also reported bootstrap confidence intervals and a permutation test for the patient-level ROC-AUC summary. All main study claims use patient-level metrics. Window-level scores appear only as intermediate outputs.

To characterize biomarker behavior across leave-one-participant-out folds, we used several post hoc summaries: mean pairwise Jaccard overlap, Kuncheva index, unique feature count, top-feature share, class-conditional selection deltas, and effect-direction consistency. These summaries are descriptive. They characterize recurrence and directionality across folds and do not support causal interpretation. Low overlap can coexist with recurring features that retain consistent class-separation direction when selected [? ? ?].

3 Results

3.1 Sweep-level comparison

Across the 150 successful runs (75 hybrid, 75 SVM), the hybrid model outperformed the SVM baseline on patient-level discrimination metrics over most of the sweep landscape. The performance advantage was visible when averaging over window sizes (Figure 3) and when averaging over **inner-k** settings (Figure 4). The best individual hybrid run occurred at 6 s windows with **inner-k=1**, whereas the best SVM run occurred at 8 s windows with **inner-k=30**.

To assess statistical robustness across the structured sweep, we applied three correction tiers to the permutation p-values. Treating the sweep as two model-family comparisons (the primary independent factor) yields Bonferroni-corrected $p_{\text{corr}} = 0.016$, confirming that the result survives at $\alpha = 0.05$. A max-T permutation correction (10,000 iterations with shared label permutations across all 150 configurations, accounting for inter-configuration correlations) yields $p_{\text{corr}} = 0.302$; the null maximum distribution has mean 0.768 and 95th percentile 0.908, placing the observed maximum of 0.816 within the range expected under random label permutations. The full-sweep

Bonferroni ($p_{\text{corr}} = 1.00$) is mechanically uninformative given the 150-test multiplier and is reported for completeness. We adopt the structured 2-family correction as the primary corrected p-value because the window-size and inner-k factors represent exploratory levels of a single hypothesis rather than independent comparisons. Across the sweep, 2 of 150 runs (1.3%) yielded raw $p < 0.05$, and 88% of runs had $p \geq 0.50$, consistent with a null-dominated landscape from which the best hybrid configuration meaningfully separates.

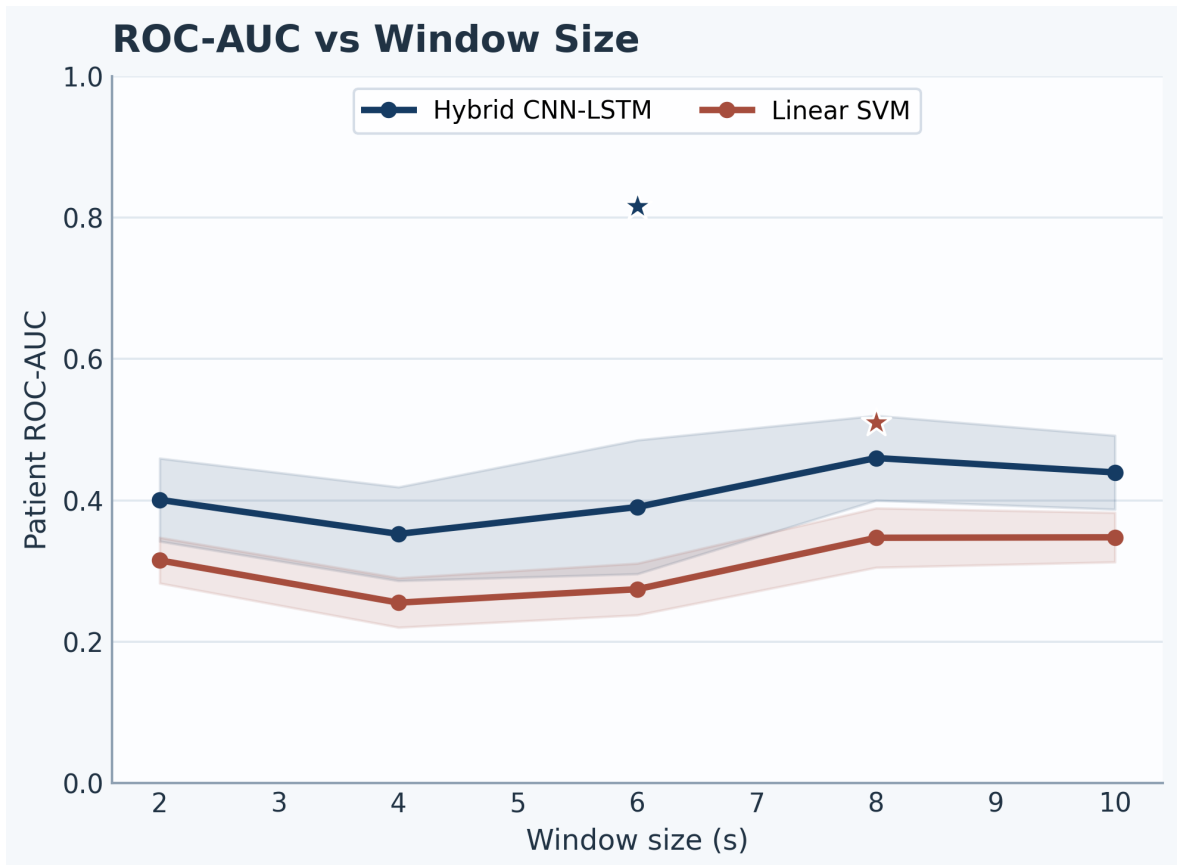


Figure 3: Patient-level ROC-AUC by window size across the sweep. Stars mark the best individual hybrid and SVM runs.

Figure 4 shows that the relationship between ROC-AUC and **inner-k** differed across the two model families in the low-budget region. The hybrid model retained clearly stronger ROC-AUC across the sparsest budgets and achieved the highest individual ROC-AUC in the entire sweep at **inner-k=1**. The linear SVM did not show the same pattern and reached its best ROC-AUC only at the broader **inner-k=30** setting. Sparse per-fold selection therefore aligned with peak discrimination only for the more expressive model.

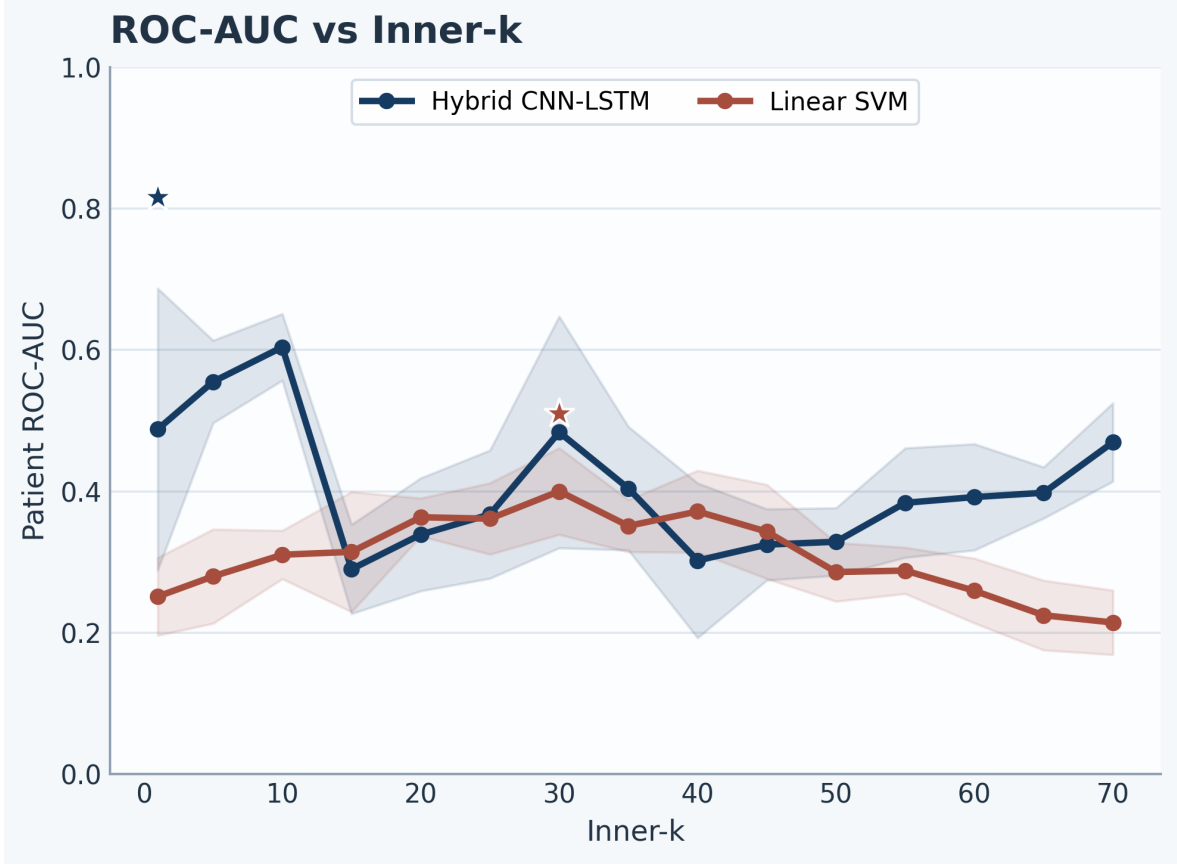


Figure 4: Patient-level ROC-AUC by `inner-k` across the sweep. The hybrid model retained the strongest discrimination in the sparsest budget region.

Figures 5 and 6 show a second related pattern. Increasing `inner-k` increased hard-set overlap and the total number of unique selected features. Larger feature budgets did not produce the strongest patient-level discrimination. The sweep concentrated its best hybrid performance at the sparsest end of the per-fold selection range. This pattern suggests that model capacity and per-fold sparsity mattered together rather than independently.

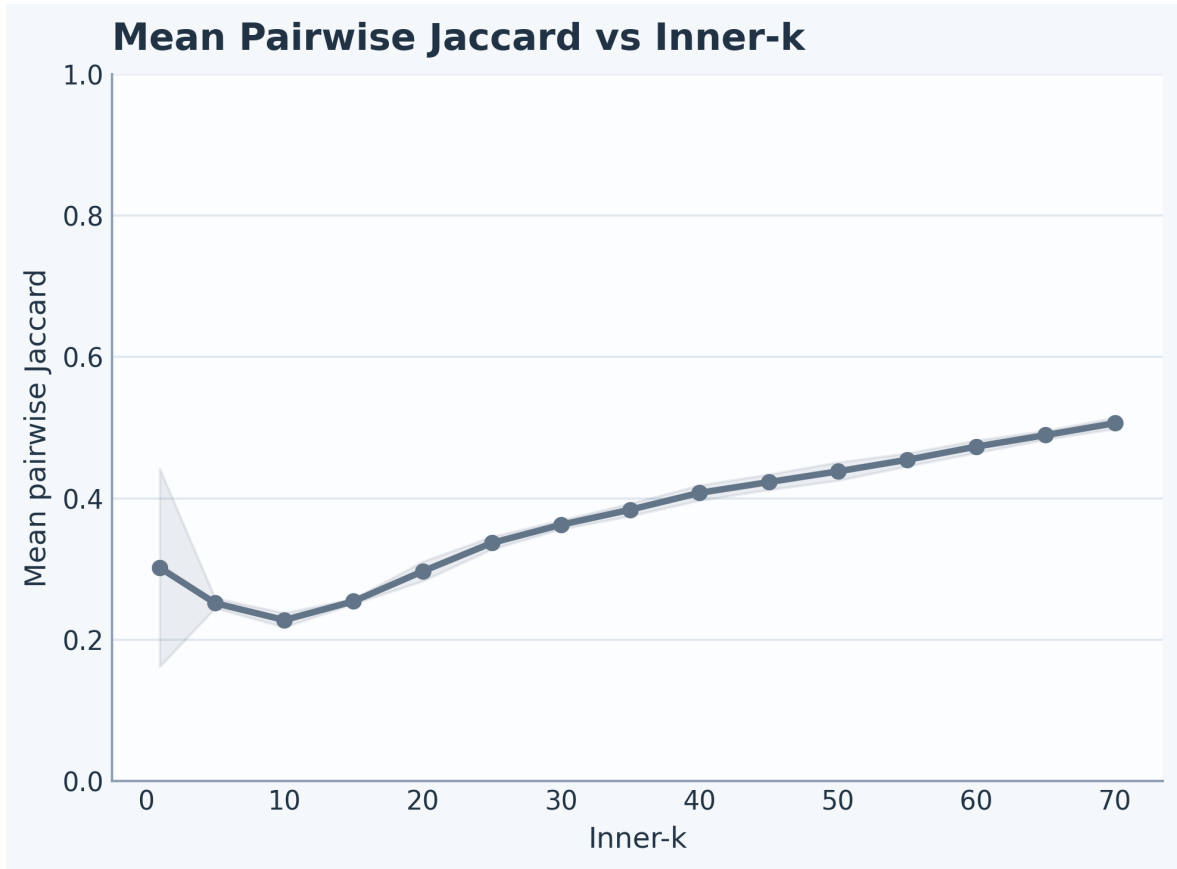


Figure 5: Mean pairwise Jaccard overlap by `inner-k`. Hard-set overlap increased with larger per-fold feature budgets.

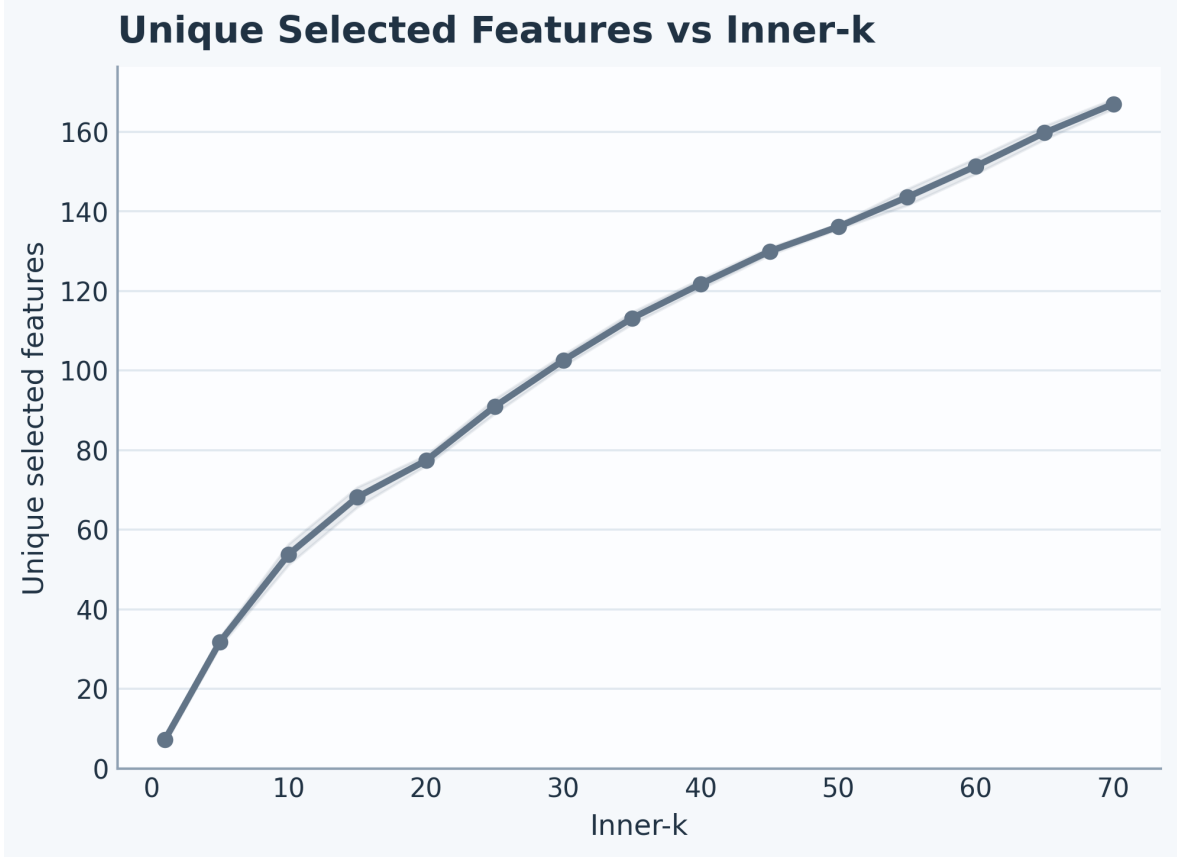


Figure 6: Unique selected features by **inner-k**. Larger per-fold budgets produced markedly larger discovered feature pools.

3.2 Best-run patient-level performance

The best hybrid configuration achieved patient-level ROC-AUC 0.816 [95% CI: 0.600–0.990], PR-AUC 0.579, balanced accuracy 0.786 [95% CI: 0.571–0.967], accuracy 0.810 [95% CI: 0.619–0.952], F1 0.714 [95% CI: 0.364–0.941], and MCC 0.571 over 21 held-out participants (Figure 7; Table 4). The permutation test yielded a nominal raw $p = 0.008$ based on 500 iterations. Correcting for the structured sweep (2 model families) gives $p_{\text{corr}} = 0.016$, which survives at $\alpha = 0.05$; under the more conservative max-T correction across all 150 configurations, $p_{\text{corr}} = 0.302$ (Table 4 footnote). Bootstrap confidence intervals are wide, reflecting the small sample size of 21 participants. The PR-AUC of 0.579 indicates that the model retained useful precision-recall behavior in this cohort.

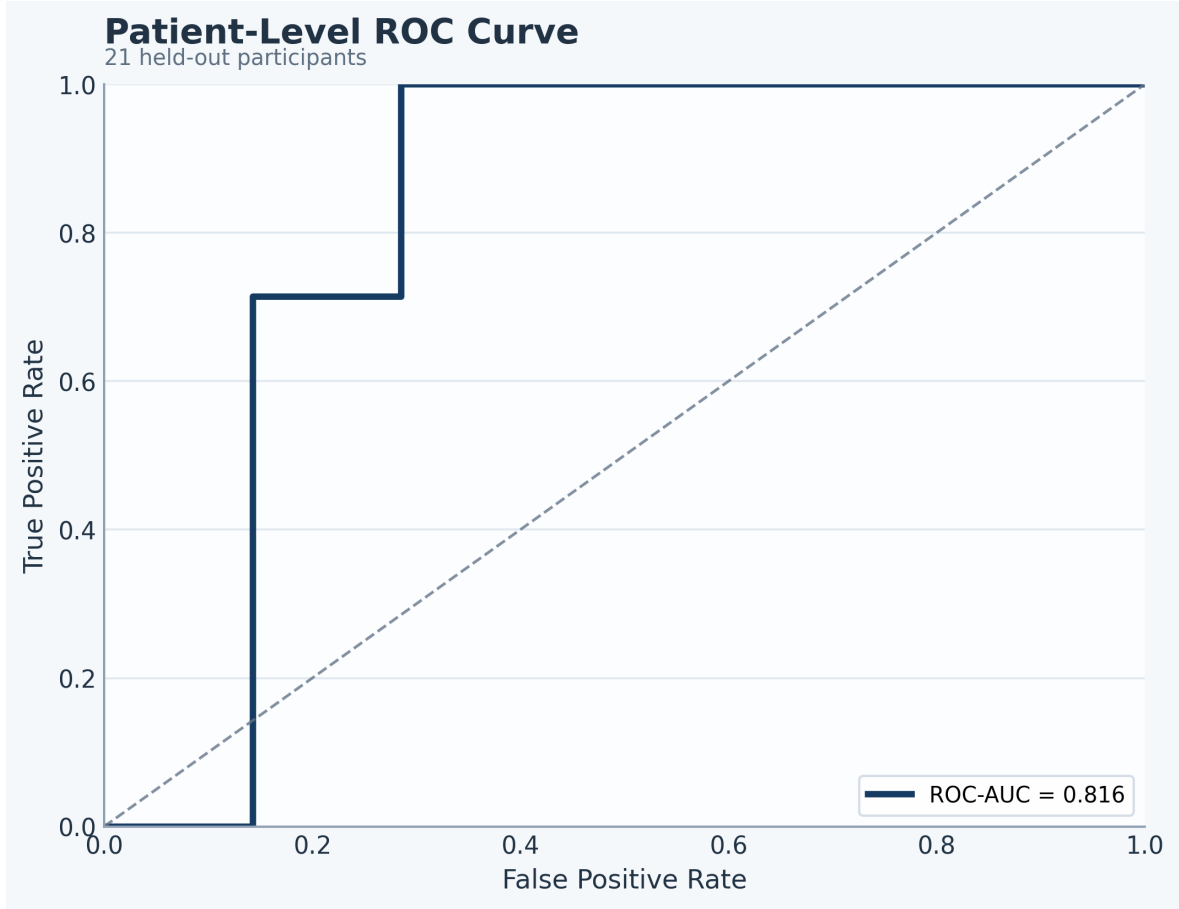


Figure 7: Patient-level ROC curve for the best hybrid run. The shaded region shows the 95% bootstrap confidence band.

The corresponding participant-level confusion counts were 12 true negatives, 5 true positives, 2 false positives, and 2 false negatives. This error profile is consistent with the balanced-accuracy summary and shows that the strongest run retained useful sensitivity without sacrificing specificity. Figure 8 shows the same run in condensed metric form with 95% bootstrap confidence intervals.

The best SVM baseline reached balanced accuracy 0.643 and ROC-AUC 0.510 [95% CI: 0.119–0.881], with a non-significant permutation result (raw $p = 0.469$, uncorrected). This weaker performance came with a much larger per-fold feature budget (**inner-k=30**) and a far larger overall candidate set. The contrast indicates that the sparse hybrid pipeline extracted stronger patient-level discrimination from a much smaller subset of features.

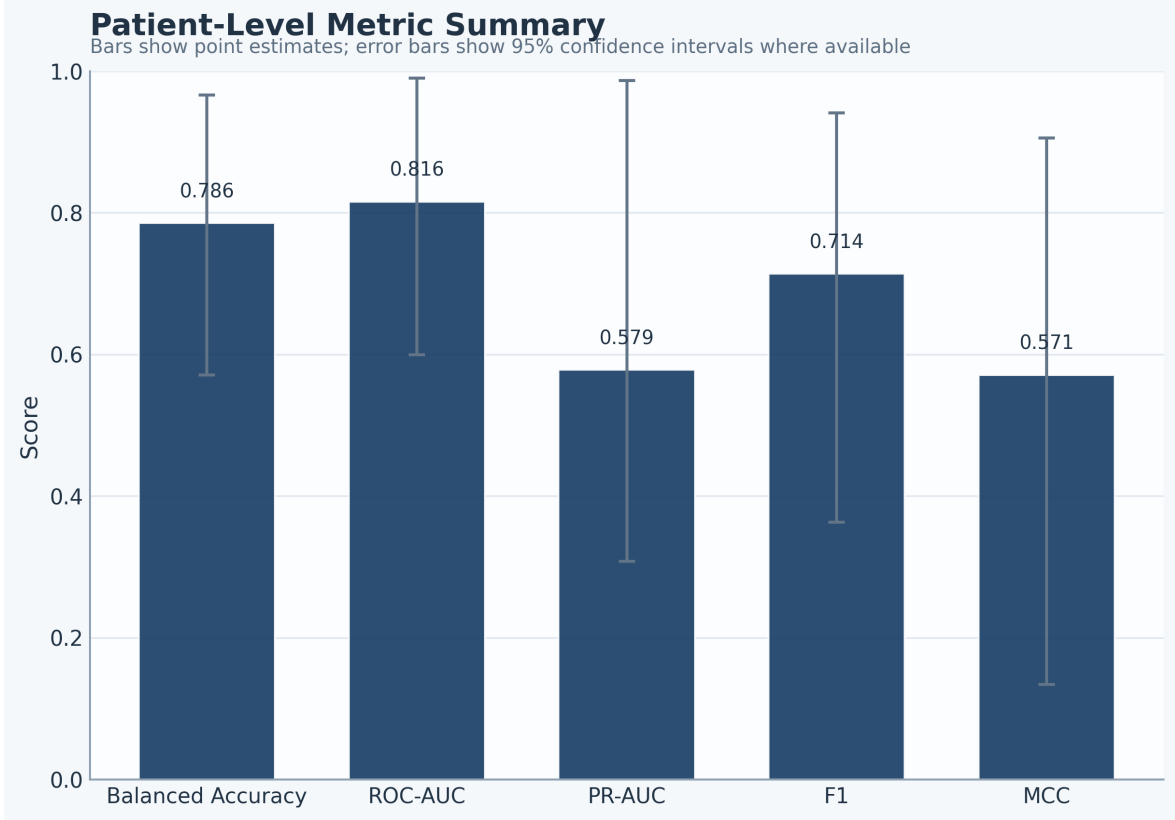


Figure 8: Patient-level metric summary for the best hybrid run. Error bars show 95% bootstrap confidence intervals.

Table 4: Best-performing configuration for each model family. Bootstrap confidence intervals are from 1,000 resamples at the 95% level.

Model	Window	Inner-k	ROC-AUC	PR-AUC	Bal. Acc.	F1	MCC	Feat./fold	Unique feat.	Mean Jaccard
Hybrid CNN-LSTM	6	1	0.816	0.579	0.786	0.714	0.571	1.0	7	0.190
Linear SVM	8	30	0.510	0.411	0.643	0.533	0.277	30.0	103	0.360

Bootstrap 95% CIs from 1,000 resamples. Hybrid permutation test: raw $p = 0.008$; primary correction (2 model families) $p_{\text{corr}} = 0.016$; robustness checks: max-T across 150 configurations $p_{\text{corr}} = 0.302$, full-sweep Bonferroni $p_{\text{corr}} = 1.00$.

3.3 Intermediate baselines at the sparsest budget

To determine whether the hybrid architecture’s advantage at **inner-k=1** reflected generic benefits of non-linearity or specific architectural properties, we evaluated eight classical classifiers at the identical configuration (6 s windows, **inner-k=1**, SMOTE, LOPO equalization, same dataset and folds). All eight performed at or below chance (Table 5). The best classical model (Random Forest) reached ROC-AUC 0.296, compared with 0.816 for the hybrid. No classical model achieved a significant permutation p-value (all $p > 0.9$). This indicates that the hybrid CNN-LSTM’s performance at extreme sparsity is not attributable to non-linearity alone; the convolutional, recurrent, and attention stages appear necessary to extract discriminatory signal from a single feature per fold.

Table 5: Intermediate baseline classifiers evaluated at the best hybrid configuration (6 s, **inner-k=1**). All models used identical data, feature selection, and validation splits.

Model	ROC-AUC	Bal. Acc.	MCC	F1	p (perm.)
Hybrid CNN-LSTM (best)	0.816	0.786	0.571	0.714	0.008
Random Forest	0.296	0.321	-0.337	0.222	0.930
Extra Trees	0.286	0.500	0.000	0.375	0.934
AdaBoost	0.286	0.393	-0.204	0.250	0.942
Gradient Boosting	0.276	0.393	-0.204	0.250	0.936
Decision Tree	0.276	0.500	0.000	0.308	0.946
KNN	0.255	0.464	-0.067	0.353	0.974
SVM-RBF	0.204	0.429	-0.135	0.333	0.978
SVM-Linear	0.204	0.357	-0.277	0.300	0.988

All models evaluated with the same configuration: window size 6 s, **inner-k=1**, SelectKBest (F -classif), SMOTE enabled, LOPO group equalization enabled, patient-level mean aggregation. Permutation p -values from 500 iterations. The hybrid result is shown for reference (not part of this sweep).

The winning hybrid run selected an average of 1 feature per fold and yielded only 7 unique features that appeared in at least one fold across the entire participant-level validation process (Table 6). That degree of sparsity is one of the most distinctive findings in the sweep. Hard-set overlap was modest, with a mean pairwise Jaccard value of 0.190 and a mean Kuncheva index of 0.187. Figure 9 shows the small candidate pool, while the overlap statistics indicate that recurrence was concentrated in a few repeatedly reappearing features rather than in a broadly shared foldwise set.

A small set of features emerged repeatedly, though not in every fold, and the selected features were dominated by temporal and frontal-temporal descriptors. The most frequently selected feature was `tp10_spectral_entropy`, selected in 8 of 21 folds. The next most frequent were `left_frontal_temporal_diff_gamma`, `left_frontal_temporal_diff_beta`, and `af7_zero_crossings`. We note that three of the seven features listed in Table 6 appear in only 1 of 21 folds; the term “recurrent” should be interpreted with this sparsity in mind, and the sign-consistency values for these single-fold features carry limited interpretive weight.

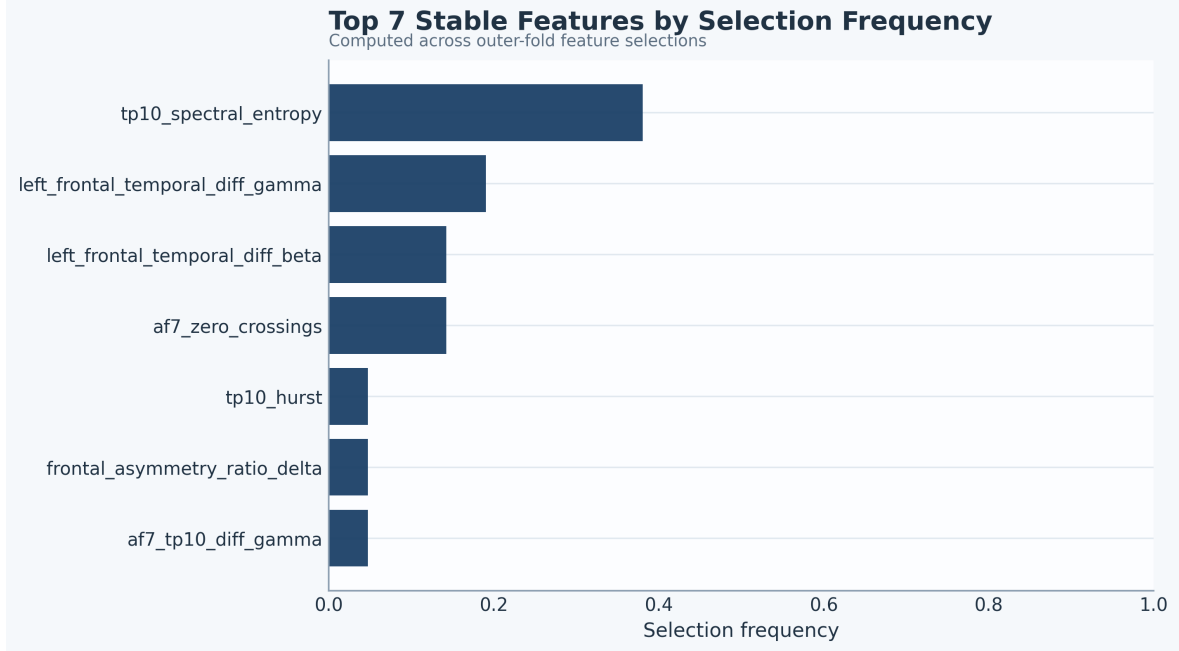


Figure 9: Selection-frequency ranking for the features identified from the best hybrid run.

Table 6: Features from the best hybrid run that appeared in at least one fold. Δ selection is remission minus non-remission selection share. Sign consistency is 1.000 for features appearing in multiple folds; single-fold entries should be interpreted with caution.

Feature	Folds	Share	Δ sel.	Sign cons.	Mean d	Effect direction
tp10_spectral_entropy	8	0.381	-0.571	1.000	-0.935	Non-remission > remission
left_frontal_temporal_diff_gamma	4	0.190	-0.071	1.000	-1.059	Non-remission > remission
left_frontal_temporal_diff_beta	3	0.143	0.000	1.000	-0.915	Non-remission > remission
af7_zero_crossings	3	0.143	0.429	1.000	0.814	Remission > non-remission
tp10_hurst	1	0.048	0.143	1.000	-1.010	Non-remission > remission
af7_tp10_diff_gamma	1	0.048	0.143	1.000	-1.203	Non-remission > remission
frontal_asymmetry_ratio_delta	1	0.048	-0.071	1.000	0.794	Remission > non-remission

3.4 Effect-direction consistency and class-conditional patterns

The recurrent hybrid biomarkers showed perfect mean sign consistency, with all recurrent features retaining the same effect direction across the folds in which they were selected. **tp10_spectral_entropy** and the frontal-temporal gamma and beta difference terms were repeatedly associated with the non-remission group, whereas **af7_zero_crossings** was more frequently selected in remission folds. The class-conditional selection frequencies make that separation visible at the level of foldwise recurrence (Figure 10).

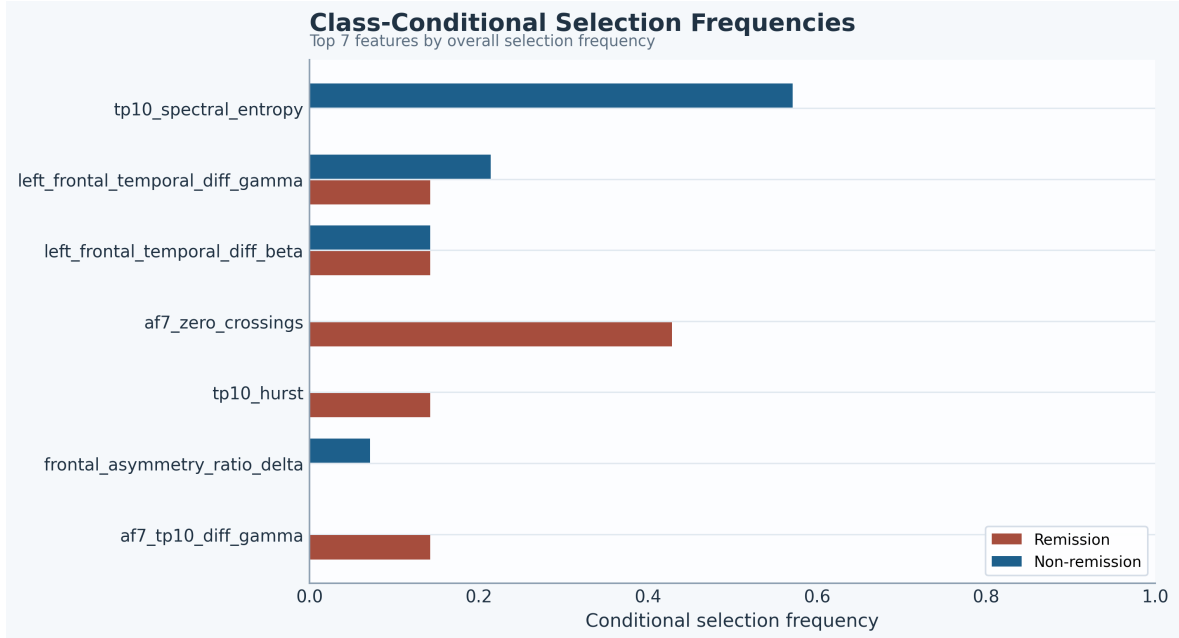


Figure 10: Class-conditional selection frequencies for the recurrent biomarkers from the best hybrid run. Remission is shown in red and non-remission in blue.

The effect-size summaries followed the same broad pattern, with **af7_zero_crossings** retaining a positive mean effect size and the main TP10 and frontal-temporal features retaining negative mean effects (Figure 11). This distinction matters clinically. The selection-frequency deltas do not justify definitive mechanistic claims. They provide a concrete interpretation target for follow-up validation. Taken together, these results indicate that useful patient-level discrimination can be achieved with an extremely sparse feature budget, while the recurrent biomarkers retain interpretable directionality even when fold-to-fold set overlap remains modest.

3.5 Aggregation robustness

Patient-level metrics in the main analysis were derived by averaging window probabilities within each participant. To assess sensitivity to outlier windows, we recomputed the best hybrid run’s patient-level metrics using median aggregation and 25% trimmed-mean aggregation. Median aggregation yielded ROC-AUC 0.857, PR-AUC 0.620, and MCC 0.571; trimmed-mean aggregation yielded ROC-AUC 0.837 and MCC 0.571. Both alternative aggregation methods matched or exceeded the original mean-aggregated performance (ROC-AUC 0.816), confirming that the main results are not driven by outlier windows and that the reported mean-based metrics are conservative rather than inflated.

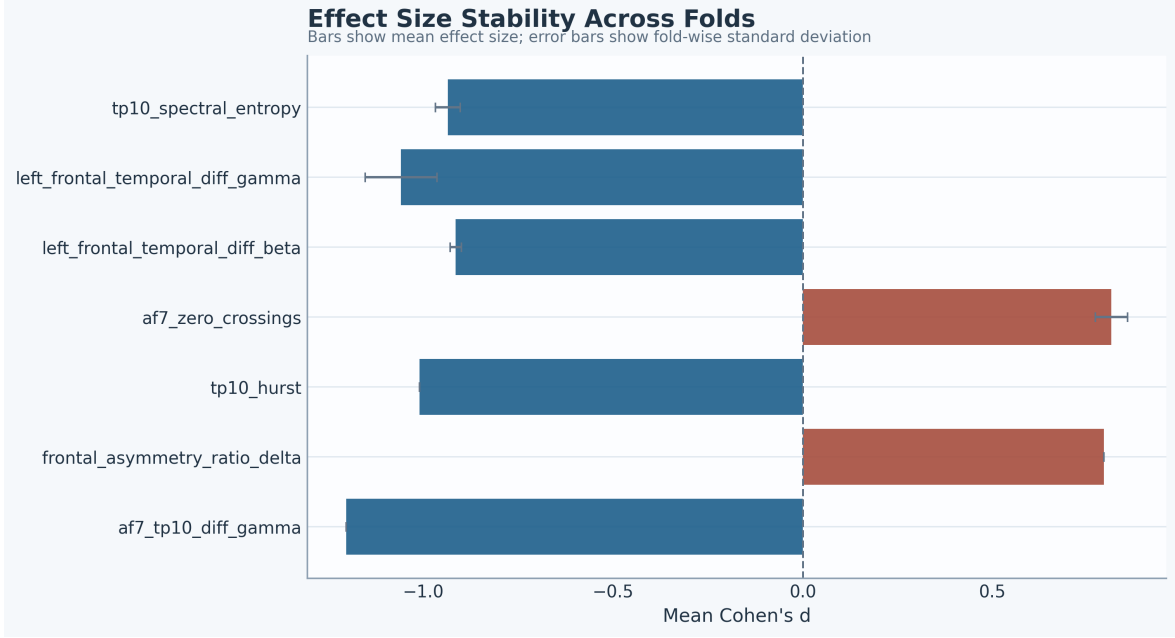


Figure 11: Effect-size stability for the recurrent biomarkers from the best hybrid run. Positive values indicate remission-skewed effects and negative values indicate non-remission-skewed effects.

4 Discussion

The strongest patient-level discrimination in this sweep emerged from the sparsest tested configuration. The best hybrid run used `inner-k=1`, selected only 7 unique features appearing in at least one fold, and exceeded the best linear baseline by a large margin on ROC-AUC and MCC. Figure 4 also shows that this was not a generic effect of smaller feature budgets across both models. In the low-budget region, the hybrid model retained stronger ROC-AUC than the linear SVM and reached its peak performance at the sparsest tested setting, whereas the linear model reached its best ROC-AUC only after a broader per-fold budget. In this cohort, the value of the more expressive model and the value of strong sparsity appear to have operated together.

Statistical inference was assessed with permutation tests and three tiers of multiple-comparisons correction to account for the structured sweep design. The raw permutation p-value of $p = 0.008$ survives the primary structured correction (2 model families, $p_{\text{corr}} = 0.016$) at $\alpha = 0.05$. As robustness checks, we computed a max-T permutation correction across all 150 configurations ($p_{\text{corr}} = 0.302$) and a full-sweep Bonferroni ($p_{\text{corr}} = 1.00$), the latter being mechanically uninformative given the 150-test multiplier. We adopt the 2-family correction as the primary inference because the inner-k and window-size factors represent exploratory levels of a single hypothesis rather than independent comparisons, and report all tiers for complete transparency. Several additional caveats qualify the results. The study is limited to $n = 21$ participants with a 7:14 class imbalance, producing wide bootstrap confidence intervals (e.g., ROC-AUC 95% CI: 0.600–0.990) that span both chance-level and near-perfect performance. The hybrid model’s hyperparameters were inherited from a standard configuration rather than tuned specifically on this cohort; while this reduces researcher degrees of freedom, the configuration may not be optimal for the present data.

One plausible explanation for the performance gap between models is that aggressive per-fold pruning removed weak or redundant handcrafted features and exposed a compact signal set whose

interactions were more useful to the hybrid architecture than to a linear decision boundary. Supporting this interpretation, eight additional classical classifiers (Random Forest, Gradient Boosting, Extra Trees, SVM-RBF, KNN, Decision Tree, AdaBoost, and SVM-Linear) were evaluated at the identical **inner-k=1** configuration with the same data, feature selection, and validation splits; all performed at or below chance (ROC-AUC 0.204–0.296, Table 5), confirming that the hybrid’s advantage is not attributable to generic non-linearity. With only a few retained descriptors, the convolutional, recurrent, and attention stages appear better positioned to combine complementary temporal and frontal-temporal information without dilution from a larger pool of unstable candidates. The linear baseline appears to have depended more on accumulating a broader set of moderate additive cues before reaching its best ROC-AUC. We did not compute formal statistical comparisons (e.g., DeLong test) between model families because the two best configurations operate at different inner-k values (1 vs. 30), making direct paired comparison ambiguous; such comparisons would be appropriate in a follow-up study with matched configurations. This interpretation remains provisional, but it is consistent with the sweep pattern and with the marked separation between the best sparse hybrid run and the broader-budget linear baseline.

This result fits part of the EEG depression literature and supports a measured interpretation. Prior work has argued that EEG can provide clinically useful baseline markers in major depressive disorder and treatment-response settings [? ? ?]. Related discussions have also emphasized tighter biomarker discipline, better standardization, and more realistic expectations about generalization [? ? ?]. Machine learning studies have likewise reported encouraging treatment-response signals, including combinations of EEG and clinical covariates [? ? ?]. Other modelling studies reach similar conclusions while still exposing limits from data heterogeneity, cohort size, or insufficient interpretability [? ? ?]. Recent predictive overviews and reviews continue to highlight the same translational tension [? ?]. Table 7 places the present results in the context of comparable EEG-based treatment-response prediction studies. The achieved ROC-AUC of 0.816 is competitive with prior work using larger montages and cohorts, despite the present study’s substantially reduced channel count and sparse feature budget. However, direct numerical comparison is complicated by differences in clinical populations, outcome definitions, and validation methodology; the table is provided for contextual orientation rather than formal benchmarking.

Table 7: Comparison with selected EEG-based treatment-response prediction studies in depression.

Study	Method	EEG channels	n	AUC	Ext. val.	Task
[? ?]	Deep learning (PSD features)	4 (AF7,AF8,TP9,TP10)	21	0.76 [†]	No	tDCS re
[? ?]	PLV connectivity + ML	4 (AF7,AF8,TP9,TP10)	21	0.72 [†]	No	tDCS re
[? ?]	SVM, RF, LR (clinical + EEG)	19	80	0.82	No	Antidepr
[? ?]	SVM, LR (EEG + clinical)	32	73	0.83	No	Antidepr
[? ?]	LORETA + SVM	72	65	0.75	No	Sertraline
[? ?]	Nonlinear EEG + classifier	19	30	0.90	No	Antidepr
[? ?]	Nonlinear features + SVM	19	45	0.88	No	rTMS re
[? ?]	SVM with spectral features	19	44	0.76	No	Treatme
This study	Sparse CNN-LSTM + handcrafted features	4 (AF7,AF8,TP9,TP10)	21	0.82	No	tDCS re

[†]Estimated from reported metrics. AUC = area under the ROC curve (patient-level where available). Ext. val. = external validation cohort. Studies selected for relevance to EEG-based treatment response prediction in depression; the list is illustrative rather than exhaustive.

The present sweep adds a feature-selection view in which larger feature budgets yield more overlap and many more unique selected features. Those larger budgets do not improve discrimination. The panels support a small set of frequently selected features with consistent

direction of class separation. That reading keeps the modest Jaccard and Kuncheva scores in view and gives appropriate weight to the stronger effect-direction signal. However, the low Jaccard (0.190) and Kuncheva (0.187) scores indicate that the selected feature changes in most folds, which is a limitation for biomarker reproducibility claims. Three of the seven features appear in only a single fold, and the “perfect” sign consistency values for those entries should be interpreted cautiously. The Jaccard increase with larger inner- k is partly mechanical (selecting k features from a fixed pool of 247 will produce more overlap by construction as k increases), so this trend should not be over-interpreted as evidence of underlying signal stability. Methodologically, this framing matters because the depression EEG biomarker literature still contains substantial publication bias, limited out-of-sample validation, and known vulnerability to leakage from pre-cross-validation feature processing [? ?]. The present pipeline addresses that problem by embedding selection and rebalancing inside participant-level leave-one-participant-out folds and then summarizing recurrence and directionality instead of reporting only a single global feature list [?].

The same clinical program has also supported other signal representations in recent PSD-based and PLV-based studies, which located informative response-related signal in low-density temporal and frontal-temporal patterns [? ?]. In the present sweep, the most frequently selected features again concentrated in TP10-derived and frontal-temporal descriptors. That regional convergence supports the view that the low-montage recordings may contain response-relevant information across multiple feature families. However, the current study provides only within-cohort evidence; the generalizability of these features under different montage configurations, clinical populations, and treatment protocols remains untested.

The specific features selected in the present sweep are neurophysiologically plausible. Elevated temporal spectral entropy in non-remission participants suggests greater irregularity or complexity in the resting-state signal at the TP10 site, which overlies the temporoparietal junction — a region implicated in the default mode network and self-referential processing, both of which are dysregulated in depression. Higher entropy may reflect less organized or more variable neural dynamics that are less amenable to tDCS-induced modulation. The frontal-temporal beta and gamma difference features point toward inter-regional communication in higher frequency bands, which have been linked to top-down cognitive control and attentional processes mediated by fronto-temporal networks. tDCS applied over the dorsolateral prefrontal cortex (approximated by AF7/AF8) is thought to modulate excitability in these circuits, and baseline differences in frontal-temporal coupling may index a network’s capacity to respond to such modulation. The AF7 zero-crossing count — a time-domain measure of signal volatility — showed the clearest remission-associated pattern and may capture a more dynamic or responsive frontal signal state that is conducive to stimulation-induced plasticity. These interpretations are necessarily speculative given the study’s sample size and lack of independent replication, but they provide concrete, testable hypotheses for future mechanistic work.

Related tDCS studies in bipolar depression from the same group point in a similar low-montage direction. PSD-based modeling again highlighted AF7 and TP10 signal content, and PLV analyses emphasized frontal-temporal and temporoparietal synchronization patterns [? ?]. These bipolar studies address a different diagnosis and treatment protocol. They support reduced-montage feasibility and signal plausibility in adjacent affective cohorts.

Several limitations of this study should be noted. First, the relatively small participant size ($n = 21$) with a 7:14 class imbalance limits statistical power and generalizability; the wide bootstrap confidence intervals (e.g., ROC-AUC 95% CI: 0.600–0.990) reflect this uncertainty. Although SMOTE and group equalization were applied within training folds, the fundamental class imbalance remains a constraint. Second, the primary corrected p-value ($p_{\text{corr}} = 0.016$) survives the structured 2-family Bonferroni correction, but does not survive the more conservative

max-T correction across all 150 configurations ($p_{\text{corr}} = 0.302$); replication in an independent cohort is needed to establish robustness. Third, there is no external validation cohort, so the reported performance and identified features cannot yet be assessed for generalizability to independent populations, alternative low-density montages, or different clinical settings. Fourth, important clinical variables such as medication status, illness duration, episode count, and psychiatric comorbidities were not available; these may influence both EEG features and treatment outcomes. Fifth, the low-montage 4-channel setup captures only frontal and temporal regions, missing spatial information from central, parietal, and occipital sites. Sixth, the feature stability metrics (mean Jaccard 0.190, Kuncheva 0.187) indicate limited cross-fold consistency; however, median-based patient aggregation improved ROC-AUC to 0.857, suggesting that the feature signal is robust even when fold-to-fold feature identity varies. Seventh, the SOTA comparison (Table 7) is contextual rather than a formal benchmark due to differences in clinical populations, outcome definitions, and validation methodology between studies; direct method-level comparisons would require harmonized datasets and protocols. Eighth, the study does not benchmark against recent transformer-based or graph-neural-network EEG classifiers, which represent an active area of development. Finally, the computational requirements of the hybrid architecture (approximately 1.2 million parameters, GPU required for training) may constrain deployment in some settings. Eighth, when `inner-k=1`, SMOTE operates in a 1-dimensional feature space where the standard motivation for synthetic minority oversampling does not fully apply; the contribution of SMOTE to the performance at this setting warrants dedicated ablation. Finally, the study does not compare against recent transformer-based or graph-neural-network methods for EEG classification, which would provide additional context for the achieved performance.

The recurring features can be treated as plausible candidate features from this cohort that now warrant follow-up validation rather than as validated clinical biomarkers. Future work should test whether the same sparse feature set recurs under independent cohorts, alternative low-density montages, and explicit external validation [?]. Adjacent response-prediction work in depression and neuromodulation provides useful targets for that next stage [? ? ?]. Home-based and transport-oriented studies offer one clear direction for follow-up [? ? ?]. More recent response-signature studies also provide relevant benchmarks for that next phase [? ? ?]. Related protocol and review work in treatment-resistant depression can help frame that validation effort [? ?]. These extensions should keep stringent patient-level evaluation and sparse interpretable feature budgets in the core study design, ideally with pre-registered analysis plans to limit researcher degrees of freedom.

5 Conclusion

This study demonstrates that low-montage resting-state EEG can support patient-level prediction of remission outcomes in a home-based tDCS trial using an extremely sparse feature-selection framework. The strongest model achieved promising discrimination (ROC-AUC 0.816, MCC 0.571) while selecting only a small set of features, though the result does not survive correction for the full sweep of 150 configurations and should be considered exploratory.

Rather than relying on high-dimensional feature spaces, the results suggest that compact feature candidates centered on temporal entropy and frontal-temporal interactions may capture clinically relevant signal. These findings extend prior PSD-based analyses in the same cohort and support a convergent regional signal interpretation. However, the limited sample size ($n = 21$), absence of an external validation cohort, and modest feature stability (Jaccard 0.190) indicate that these features should be treated as hypothesis-generating rather than as validated clinical biomarkers. Prospective

validation in independent cohorts with pre-registered analysis plans is necessary before any clinical application can be considered.

CRediT authorship contribution statement

Abin Daniel Zorto: Writing - review & editing, writing - original draft, visualization, validation, software, methodology, formal analysis, data curation, conceptualization.

Jijomon C. Moncy: Writing - review & editing, writing - original draft, visualization, validation, software, resources, methodology, formal analysis, data curation, conceptualization.

Wenyi Xiao: Writing - review & editing, writing - original draft, visualization, validation, software, resources, methodology, formal analysis, data curation, conceptualization.

Mhd Saeed Sharif: Writing - review & editing, writing - original draft, validation, supervision, resources, project administration, methodology, investigation, funding acquisition, formal analysis, conceptualization.

Cynthia H.Y. Fu: Writing - review & editing, writing - original draft, validation, supervision, resources, project administration, methodology, investigation, funding acquisition, formal analysis, conceptualization.

Acknowledgements

The authors express their gratitude to the participants who took part in the studies. This publication is based on research supported by grants from Flow Neuroscience (FL001), NIMH (R01MH134236), National Institute for Health and Care Research (NIHR) Health Technology Assessment (NIHR165425), NIHR Efficacy and Mechanisms Evaluation (NIHR167361), and Rosetrees Trust (CF2024M2116, CF20212104) to C.H.Y.F.

References

- [1] R. Woodham, R. M. Rimmer, J. Mutz, and C. H. Y. Fu, “Is tdc a potential first line treatment for major depression?” *International Review of Psychiatry*, vol. 33, no. 3, pp. 250–265, may 2021.
- [2] R. D. Woodham, S. Selvaraj, N. Lajmi, H. Hobday, G. Sheehan, A.-R. Ghazi-Noori, P. J. Lagerberg, M. Rizvi, S. S. Kwon, P. Orhii *et al.*, “Home-based transcranial direct current stimulation treatment for major depressive disorder: a fully remote phase 2 randomized sham-controlled trial,” *Nature medicine*, vol. 31, no. 1, pp. 87–95, 2025.
- [3] E. Hampsey, M. Kalfas, L. Carter, M. Bloomfield, H. Rezaei, A. H. Young, and C. H. Y. Fu, “A systematic review of transcranial electrical stimulation and meta-analysis of transcranial direct current stimulation rcts in unipolar and bipolar depression,” *Journal of Affective Disorders*, vol. 400, p. 121009, may 2026.
- [4] R. M. Rimmer, R. D. Woodham, S. Cahill, and C. H. Fu, “Acceptability of home-based transcranial direct current stimulation (tdcs) in major depression: a qualitative analysis of individual experiences,” *Mental Health Review Journal*, vol. 29, no. 1, pp. 79–91, 2024.
- [5] S. Olbrich and M. Arns, “Eeg biomarkers in major depressive disorder: Discriminative power and prediction of treatment response,” *International Review of Psychiatry*, 2013. [Online]. Available: <https://doi.org/10.3109/09540261.2013.816269>

- [] S. Olbrich, R. v. Dinteren, and M. Arns, "Personalized medicine: Review and perspectives of promising baseline eeg biomarkers in major depressive disorder and attention deficit hyperactivity disorder," *Neuropsychobiology*, 2015. [Online]. Available: <https://doi.org/10.1159/000437435>
- [] A. F. Leuchter, I. A. Cook, S. P. Hamilton, K. L. Narr, A. Toga, A. M. Hunter, K. Faull, J. Whitelegge, A. M. Andrews, J. Loo, B. Way, S. F. Nelson, S. Horvath, and B. D. Lebowitz, "Biomarkers to predict antidepressant response," *Current Psychiatry Reports*, 2010. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/20963521/>
- [] C. H. Fu and S. G. Costafreda, "Neuroimaging-based biomarkers in psychiatry: clinical opportunities of a paradigm shift," *The Canadian Journal of Psychiatry*, vol. 58, no. 9, pp. 499–508, 2013.
- [] C. H. Fu, Y. Fan, and C. Davatzikos, "Addressing heterogeneity (and homogeneity) in treatment mechanisms in depression and the potential to develop diagnostic and predictive biomarkers," *NeuroImage: Clinical*, vol. 24, p. 101997, 2019.
- [] A. S. Widge, M. T. Bilge, R. Montana, W. Chang, C. I. Rodriguez, T. Deckersbach, L. L. Carpenter, N. H. Kalin, and C. B. Nemeroff, "Electroencephalographic biomarkers for treatment response prediction in major depressive illness: A meta-analysis," *American Journal of Psychiatry*, vol. 176, no. 1, pp. 44–56, 2019.
- [] L. Simmatis, E. E. Russo, J. Geraci, I. E. Harmsen, and N. Samuel, "Technical and clinical considerations for electroencephalography-based biomarkers for major depressive disorder," *npj Mental Health Research*, 2023. [Online]. Available: <https://doi.org/10.1038/s44184-023-00038-7>
- [] D. Watts, R. F. Pulice, J. Reilly, A. R. Brunoni, F. Kapczinski, and I. C. Passos, "Predicting treatment response using eeg in major depressive disorder: A machine-learning meta-analysis," *Translational psychiatry*, vol. 12, no. 1, p. 332, 2022.
- [] S. Cohen, J. Zantvoord, B. Wezenberg, J. Daams, C. Bockting, D. Denys, and G. van Wingen, "Electroencephalography for predicting antidepressant treatment success: a systematic review and meta-analysis," *Journal of Affective Disorders*, vol. 321, pp. 201–207, 2023.
- [] A. M. Al-Kaysi, A. Al-Ani, C. K. Loo, T. Y. Powell, D. M. Martin, M. Breakspear, and T. W. Boonstra, "Predicting treatment outcomes of patients with major depressive disorder using automated eeg classification," *Journal of affective disorders*, vol. 208, pp. 597–603, 2017.
- [] N. Jaworska, S. d. l. Salle, M. H. Ibrahim, P. Blier, and V. Knott, "Leveraging machine learning approaches for predicting antidepressant treatment response using electroencephalography (eeg) and clinical data," *Frontiers in Psychiatry*, 2019. [Online]. Available: <https://doi.org/10.3389/fpsyt.2018.00768>
- [] T. Oakley, J. Coskuner, A. Cadwallader, M. Ravan, and G. Hasey, "Eeg biomarkers to predict response to sertraline and placebo treatment in major depressive disorder," *IEEE Transactions on Biomedical Engineering*, 2022. [Online]. Available: <https://doi.org/10.1109/tbme.2022.3204861>
- [] R. Shalbaf, C. A. Brenner, C. Pang, D. M. Blumberger, J. Downar, Z. J. Daskalakis, J. Tham, R. W. Lam, F. Farzan, and F. Vila-Rodriguez, "Non-linear entropy analysis in eeg to predict treatment response to repetitive transcranial magnetic stimulation in depression," *Frontiers in Pharmacology*, 2018. [Online]. Available: <https://doi.org/10.3389/fphar.2018.01188>

- [] E. Ebrahimzadeh, M. Asgarinejad, S. Saliminia, S. Ashoori, and M. Seraji, "Predicting clinical response to transcranial magnetic stimulation in major depression using time-frequency eeg signal processing," *Biomedical Engineering Applications Basis and Communications*, 2021. [Online]. Available: <https://doi.org/10.4015/s1016237221500484>
- [] A. Zhdanov *et al.*, "Use of machine learning for predicting escitalopram treatment outcome from eeg in adults with depression," *Human Brain Mapping*, 2020, placeholder citation for Zhdanov SVM study.
- [] S. D. Puthankattil and P. K. Joseph, "Classification of eeg signals in normal and depression conditions by ann using rwe and signal entropy," *J. Mech. Med. Biol.*, vol. 12, no. 04, p. 1240019, Sep. 2012.
- [] S. Ghiasi, C. Dell'Acqua, S. Messerotti Benvenuti, E. P. Scilingo, C. Gentili, G. Valenza, and A. Greco, "Classifying subclinical depression using EEG spectral and connectivity measures," in *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE, Nov. 2021, pp. 2050–2053.
- [] S. Zhao, S.-C. Ng, S. Khoo, and A. Chi, "Temporal and spatial dynamics of EEG features in female college students with subclinical depression," *Int. J. Environ. Res. Public Health*, vol. 19, no. 3, Feb. 2022.
- [] M. Shim, S.-H. Lee, and H.-J. Hwang, "Inflated prediction accuracy of neuropsychiatric biomarkers caused by data leakage in feature selection," *Scientific Reports*, vol. 11, no. 1, apr 2021.
- [] A. Shen, J. Sun, X. Chen, and X. Gao, "A data-centric and interpretable eeg framework for depression severity grading using shap-based insights," *Journal of NeuroEngineering and Rehabilitation*, vol. 22, no. 1, may 2025.
- [] A. P. Funk and M. S. George, "Prefrontal eeg asymmetry as a potential biomarker of antidepressant treatment response with transcranial magnetic stimulation (tms): A case series," *Clinical EEG and Neuroscience*, 2008. [Online]. Available: <https://doi.org/10.1177/155005940803900306>
- [] N. Bailey, K. Hoy, C. Sullivan, B. Allman, N. Rogasch, Z. Daskalakis, and P. Fitzgerald, "Concurrent transcranial magnetic stimulation and electroencephalography measures are associated with antidepressant response from rtms treatment for depression," *medRxiv*, 2023. [Online]. Available: <https://www.semanticscholar.org/paper/8278c0b8d2b48ae5ed473c09d6c3f9ba34f53391>
- [] N. Murphy, A. J. F. Tamman, M. Lijffijt, D. Amarneh, S. Iqbal, A. C. Swann, L. A. Averill, B. O'Brien, and S. J. Mathew, "Neural complexity eeg biomarkers of rapid and post-rapid ketamine effects in late-life treatment-resistant depression: a randomized control trial," *Neuropsychopharmacology*, 2023. [Online]. Available: <https://doi.org/10.1038/s41386-023-01586-4>
- [] L. Guo, Z. Zhang, X. Tan, K. Phua, C. Wang, P. Tor, and K. Ang, "Resting-state eeg biomarkers of accelerated intermittent theta burst stimulation treatment for depression: a pilot study," in *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2024. [Online]. Available: <https://www.semanticscholar.org/paper/9d00f9320f2f75435719d9cccc4133e989f8737e>

- [] Y.-C. Tsai, C.-T. Li, and C.-H. Juan, “A review of critical brain oscillations in depression and the efficacy of transcranial magnetic stimulation treatment,” *Frontiers in Psychiatry*, 2023. [Online]. Available: <https://doi.org/10.3389/fpsy.2023.1073984>
- [] P. C. Dagnino, C. Braboszcz, E. Kroupi, M. Splittgerber, H. Brauer, A. Dempfle, C. Breitling-Ziegler, A. Prehn-Kristensen, K. Krauel, M. Siniatchkin *et al.*, “Stratification of responses to tdcs intervention in a healthy pediatric population based on resting-state eeg profiles,” *Scientific Reports*, vol. 13, no. 1, p. 8438, 2023.
- [] J. Sheen, F. Mazza, D. Momi, J. Miron, F. Mansouri, T. Russell, R. Zhou, M. Hyde, L. Fox, H. Voetterl, E. B. Assi, Z. Daskalakis, D. Blumberger, J. D. Griffiths, and J. Downar, “N100 as a response prediction biomarker for accelerated 1 hz right dlpc-rtms in major depression.” *Journal of Affective Disorders*, 2024. [Online]. Available: <https://www.semanticscholar.org/paper/c0dabdc70010a5fb8ea6f2546d37fe9e83a13071>
- [] A. A. Zeidabadi and S. Rashidi, “Prediction of rtms treatment response in depression using a frequency-based eeg biomarker,” in *International Conference on Computer and Knowledge Engineering*, 2025. [Online]. Available: <https://www.semanticscholar.org/paper/fa7b0c55f517ebfbaa6af5be40ba57129e5d87f2>
- [] A. Reissmann, M. Rupprecht, B. Langguth, J. Rischer, and S. Schoisswohl, “Theta-cordance as a biomarker of treatment response to intermittent theta burst stimulation in patients with treatment-resistant depression.” *Clinical Neurophysiology*, 2026. [Online]. Available: <https://www.semanticscholar.org/paper/c2c05fd78b73d7ebd82bcd5cb8af3516e1f5751>
- [] S. Ulrich, E. Schneider, G. Deuring, S. Erni, M. d. Ridder, J. Sarlon, and A. B. Bruhl, “Alterations in resting-state eeg functional connectivity in patients with major depressive disorder receiving electroconvulsive therapy: A systematic review,” *Neuroscience & Biobehavioral Reviews*, 2025. [Online]. Available: <https://doi.org/10.1016/j.neubiorev.2025.106017>
- [] J. C. Moncy, Y. Fan, and C. H. Fu, “Prediction of treatment outcome to transcranial direct current stimulation in major depression based on deep learning of eeg data,” in *2024 IEEE Conference on Artificial Intelligence (CAI)*, 2024, pp. 1123–1128.
- [] W. Xiao, J. C. Moncy, R. D. Woodham, S. Selvaraj, N. Lajmi, H. Hobday, G. Sheehan, A.-R. Ghazi-Noori, P. J. Lagerberg, R. Machado-Vieira *et al.*, “Home-based transcranial direct current stimulation (tdcs) in major depressive disorder: enhanced network synchronization with active relative to sham and deep learning-based predictors of remission,” *Personalized Medicine in Psychiatry*, vol. 49, p. 100147, 2025.
- [] P. Miranda, C. D. Cox, M. Alexander, S. Danev, and J. R. Lakey, “Overview of current diagnostic, prognostic, and therapeutic use of eeg and eeg-based markers of cognition, mental, and brain health,” *Integr. Mol. Med.*, vol. 6, no. 5, 2019.
- [] S. Siuly, Y. Li, and Y. Zhang, “Electroencephalogram (eeg) and its background,” in *EEG signal analysis and classification: Techniques and applications*. Springer, 2017, pp. 3–21.
- [] S. Mahato, N. Goyal, D. Ram, and S. Paul, “Detection of depression and scaling of severity using six channel EEG data,” *J. Med. Syst.*, vol. 44, no. 7, p. 118, May 2020.

- [] BABILONI, CLAUDIO, B. PH, B. M, D. WH, F. DH, F. J, H. U, H. K, I. P. E. G. C. Collaborators, J. M, K. T, K. VJ, L. D. S. FH, M. M, M. A, N. JF, O. S, P. LS, R. GS, S. B, S. A, S. WK, v. G. JM, W. FJ, and W. RG, "Guidelines for the recording and evaluation of pharmaco-eeg data in man: the international pharmaco-eeg society (ipeg)," 2012. [Online]. Available: https://iris.uniroma1.it/bitstream/11573/802874/1/Jobert_Guidelines_2012.pdf
- [] H. Zhang, Q.-Q. Zhou, H. Chen, X.-Q. Hu, W.-G. Li, Y. Bai, J.-X. Han, Y. Wang, Z.-H. Liang, D. Chen, F.-Y. Cong, J.-Q. Yan, and X.-L. Li, "The applied principles of eeg analysis methods in neuroscience and clinical neurology," *Military Medical Research*, vol. 10, no. 1, dec 2023.
- [] P. Welch, "The use of fast fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms," *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70–73, jun 1967.
- [] W. Mumtaz, L. Xia, M. A. Mohd Yasin, S. S. Azhar Ali, and A. S. Malik, "A wavelet-based technique to predict treatment outcome for major depressive disorder," *PLOS ONE*, vol. 12, no. 2, p. e0171409, feb 2017.
- [] N. Jaworska, H. Wang, D. M. Smith, P. Blier, V. Knott, and A. B. Protzner, "Pre-treatment eeg signal variability is associated with treatment success in depression," *NeuroImage: Clinical*, vol. 17, pp. 368–377, 2018.
- [] B. Hjorth, "Eeg analysis based on time domain properties," *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, sep 1970.
- [] O. Faust, P. C. A. Ang, S. D. Puthankattil, and P. K. Joseph, "Depression diagnosis support system based on eeg signal entropies," *Journal of Mechanics in Medicine and Biology*, vol. 14, no. 3, p. 1450035, mar 2014.
- [] U. R. Acharya, V. K. Sudarshan, H. Adeli, J. Santhosh, J. E. W. Koh, S. D. Puthankatti, and A. Adeli, "A novel depression diagnosis index using nonlinear features in eeg signals," *European Neurology*, vol. 74, no. 1-2, pp. 79–83, 2015.
- [] B. Lord and J. J. B. Allen, "Evaluating eeg complexity metrics as biomarkers for depression," *Psychophysiology*, vol. 60, no. 8, feb 2023.
- [] A. Monni, K. L. Collison, K. E. Hill, B. A. Oumeziane, and D. Foti, "The novel frontal alpha asymmetry factor and its association with depression, anxiety, and personality traits," *Psychophysiology*, vol. 59, no. 11, p. e14109, may 2022.
- [] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, jun 2002.
- [] J. C. Moncy, W. Xiao, R. D. Woodham, A.-R. Ghazi-Noori, H. Rezaei, E. Bramon, P. Ritter, M. Bauer, A. H. Young, Y. Fan, and C. H. Fu, "Deep learning based treatment remission prediction to transcranial direct current stimulation in bipolar depression using eeg power spectral density," *medRxiv*, 2025.
- [] W. Xiao, J. C. Moncy, A.-R. Ghazi-Noori, R. D. Woodham, H. Rezaei, E. Bramon, P. Ritter, M. Bauer, A. H. Young, and C. H. Fu, "Enhanced network synchronization connectivity following transcranial direct current stimulation (tdcs) in bipolar depression: effects on eeg

- oscillations and deep learning-based predictors of clinical remission,” *Journal of Affective Disorders*, vol. 369, pp. 576–587, 2025.
- [] R. Romero Marin, S. Lopez Rodriguez, S. Lakis Granell, E. Buloz Osorio, M. Cabello Toscano, M. Urretavizcaya Sarachaga, M. d. P. Alonso Ortega, M. Chopra, J. Solana Sanchez, J. Camprodon, A. Pascual Leone, D. Bartres Faz, D. Cappon, and G. Cattaneo, “Remotely supervised home-based tdcS in treatment-resistant depression: Feasibility, effectiveness and eeg biomarkers of response,” *Research Square*, 2026. [Online]. Available: <https://www.semanticscholar.org/paper/7ba628353818d20f18ac5ecdccca2bd39bca3d77>
 - [] I. H. Kratter, C. W. Austelle, J. I. Lissemore, M. Wada, A. Geoly, A. Chaiken, I. Kaloiani, N. Johnson, S. Wan, L. Kozyr, E. Makarewycz, B. L. Wong, M. Sridhar, F. M. Espil, N. Bassano, B. Kim, J. Ehrie, A. Maron Katz, C. Tischler, R. Nejad, J.-M. Batail, A. Phillips, E. Cole, T. J. Ford, B. S. Bentzley, B. Jo, A. F. Schatzberg, D. Spiegel, C. Rolle, G. L. Sahlem, and N. Williams, “Stanford neuromodulation therapy for treatment-resistant depression: a randomized controlled trial confirming efficacy, and an <scp>eeg</scp> study providing insight into mechanism of action and a potentially predictive biomarker of efficacy,” *World Psychiatry*, 2026. [Online]. Available: <https://doi.org/10.1002/wps.70032>
 - [] D. A. Waller, L. L. Carpenter, and S. R. Jones, “Transient frontal spectral events from eeg predict antidepressant response to sertraline in depression.” *Journal of Psychiatric Research*, 2026. [Online]. Available: <https://www.semanticscholar.org/paper/6a3e902c05c07569afa93b0bb3e0b484bf519824>
 - [] M. Pettoruso, G. D. Lorenzo, B. Benatti, G. dAndrea, C. Cavallotto, R. Carullo, G. Mancusi, O. D. Marco, G. Mammarella, A. Dattilio, E. Barlocchi, I. Rosa, A. Cocco, L. P. Padula, G. Bubbico, M. G. Perrucci, R. Guidotti, A. DAndrea, L. Marzetti, F. Zoratto, B. Delloso, and G. Martinotti, “Overcoming treatment-resistant depression with machine-learning based tools: a study protocol combining eeg and clinical data to personalize glutamatergic and brain stimulation interventions (selectool project),” *Frontiers in Psychiatry*, 2024. [Online]. Available: <https://www.semanticscholar.org/paper/169e53dbe54745e1df0495183343bda9d6a8a29e>